

ID:

Name:

Solution



Inspiring Excellence

Brac University

Semester: Fall 2025

Course Code: CSE251

Electronics Devices and Circuits

Set

A

Assessment: Final Exam

Duration: 1 hour 45 minutes

Date: January 13, 2026

Full Marks: 55

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the answer script and the question paper must be returned to the invigilator.
- ✓ All 5 questions are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

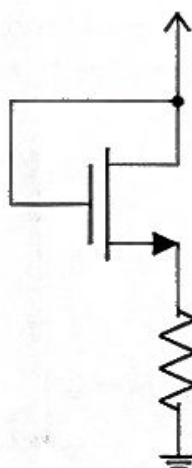
### ■ Question 1 of 5

[CO1] [8 marks]

- (a) [2 marks] The output voltages of a full-wave and a half-wave rectifier with separate filter capacitors have the same peak and ripple voltages for the same input AC sinusoidal voltage. Explain how can you identify which voltage is from which rectifier.
- (b) [3 marks] For each operating mode of an NPN BJT listed below, use a checkmark (✓) to indicate whether the base-emitter (BE) and base-collector (BC) junctions are Forward Biased (F.B.) or Reverse Biased (R.B.).

| Operating Mode | Base-Emitter (BE)                   |                                     | Base-Collector (BC)                 |                                     |
|----------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|                | F.B.                                | R.B.                                | F.B.                                | R.B.                                |
| Active         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Saturation     | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Cut-off        | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |

- (c) [3 marks] In the following circuit, given that the gate and drain terminals are short-circuited, show that if the MOSFET is ON (i.e., a gate-to-source voltage greater than the threshold voltage), it satisfies the condition to operate in the saturation region.



### ■ Question 2 of 5

[CO3] [10 marks]

- (a) [5 marks] A 12 V amplitude, 2 kHz sinusoidal AC voltage is rectified and filtered using a FW rectifier with 4 μF filter capacitor. The diodes used in the rectifier are identical and have a cut-in voltage of 1.2 V. If the average output voltage is measured to be 9 V, determine the value of the load resistance.
- (b) [5 marks] Implement the following logic function using MOSFETs. Here, A, B, C, D, and E are Boolean inputs, and f is the Boolean output:

$$f = (A + B) \cdot C + D \cdot E$$



## Set-A

1. (a) We can identify by the frequencies. If output will have same frequency, SW output will have double frequency compared to the input.

(b) In Question:

(c) For saturation

$$V_{DS} \geq V_{GS} - V_T$$

$$\Rightarrow V_D - V_S \geq V_G - V_S - V_T$$

$$\Rightarrow \underbrace{V_G - V_S}_{\text{X}} \geq \underbrace{V_G - V_S}_{\text{X}} - V_T$$

$\Rightarrow$

i.e. will always be greater than R.S.

2. (a)  $V_{dc} = V_p - \frac{V_r}{2}$

$$V_p = V_m - 2V_D$$

$$= 9.6V$$

$$V_r = 2(V_p - V_{dc})$$

$$= 2(9.6 - 9)$$

$$= 1.2V$$

$$V_r = \frac{V_p}{f_o R C}$$

$$\Rightarrow R = \frac{V_p}{f_o R V_r}$$

$$= 500\Omega$$

$$V_m = 12V$$

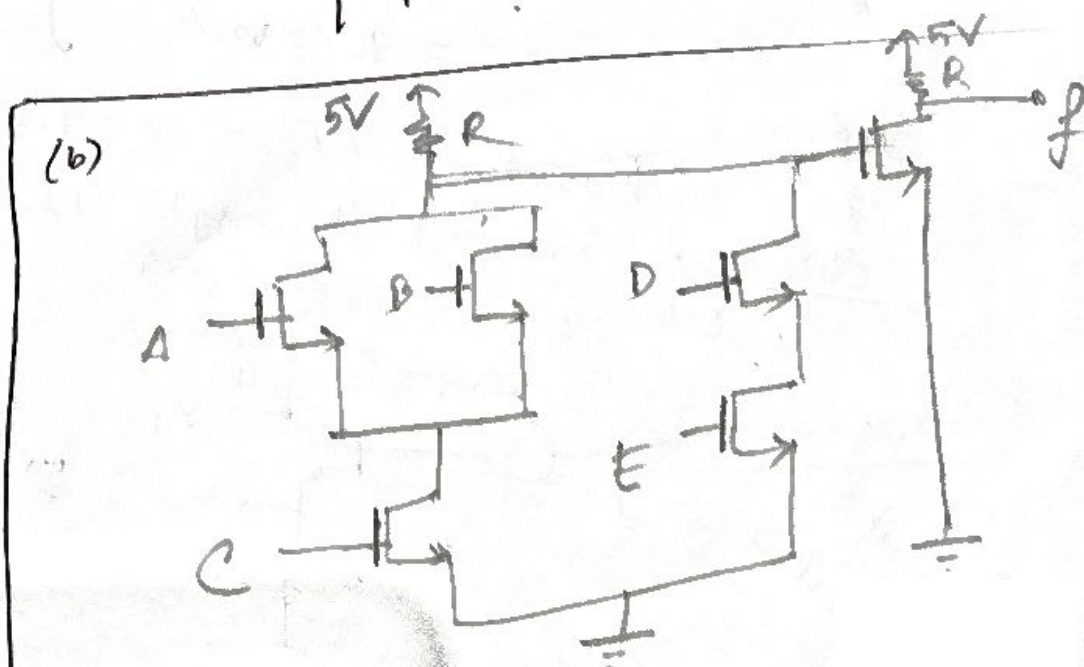
$$f_i = 2kHz, f_o = 4kHz$$

$$C = 4\mu F$$

$$V_{D0} = 1.2V$$

$$V_{dc} = 9V$$

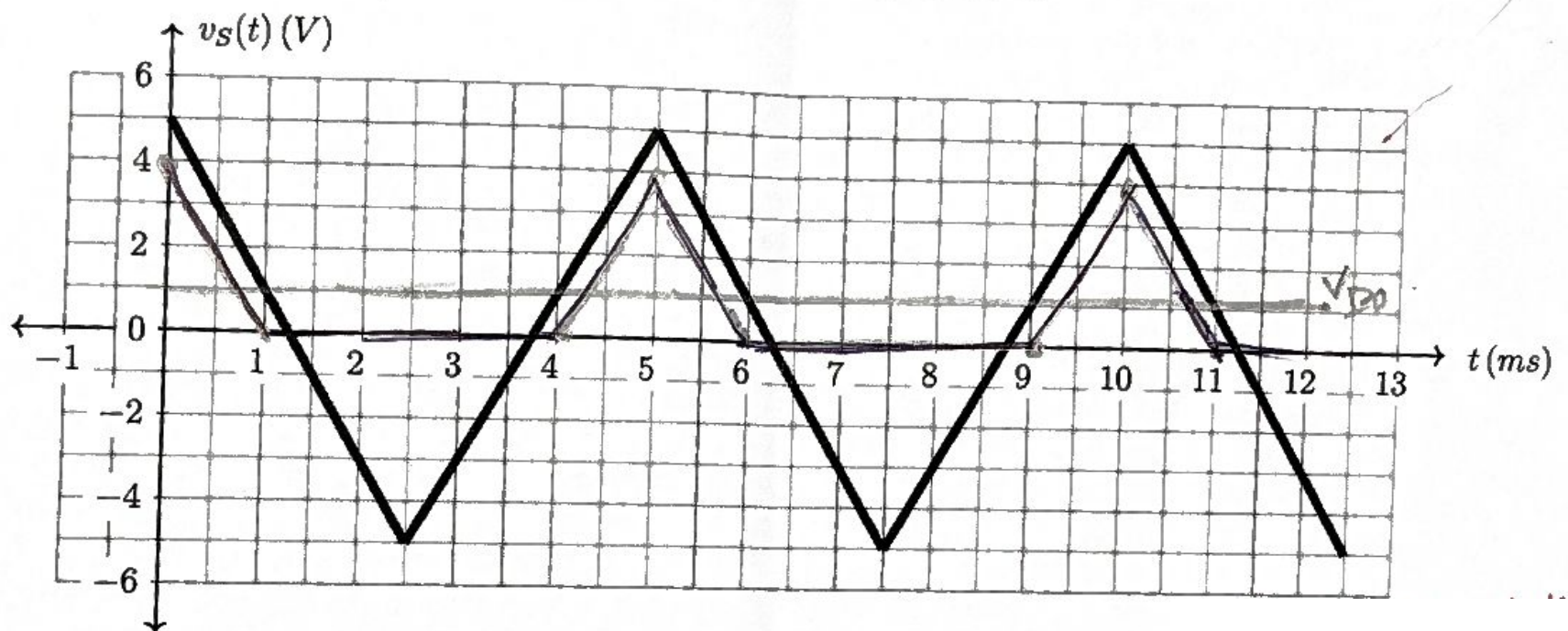
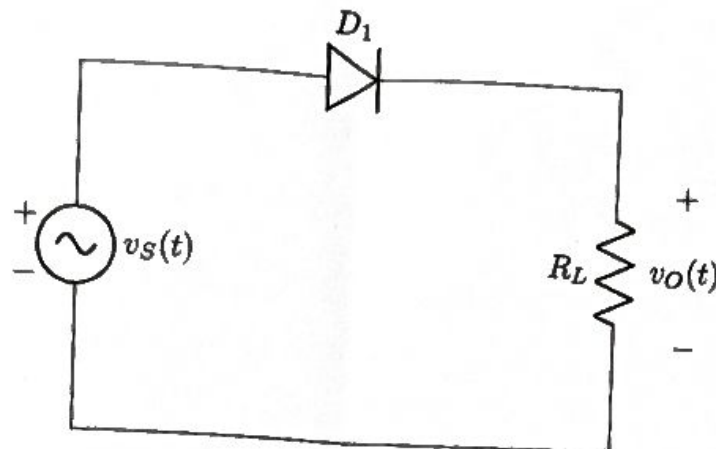
$$R = ?$$



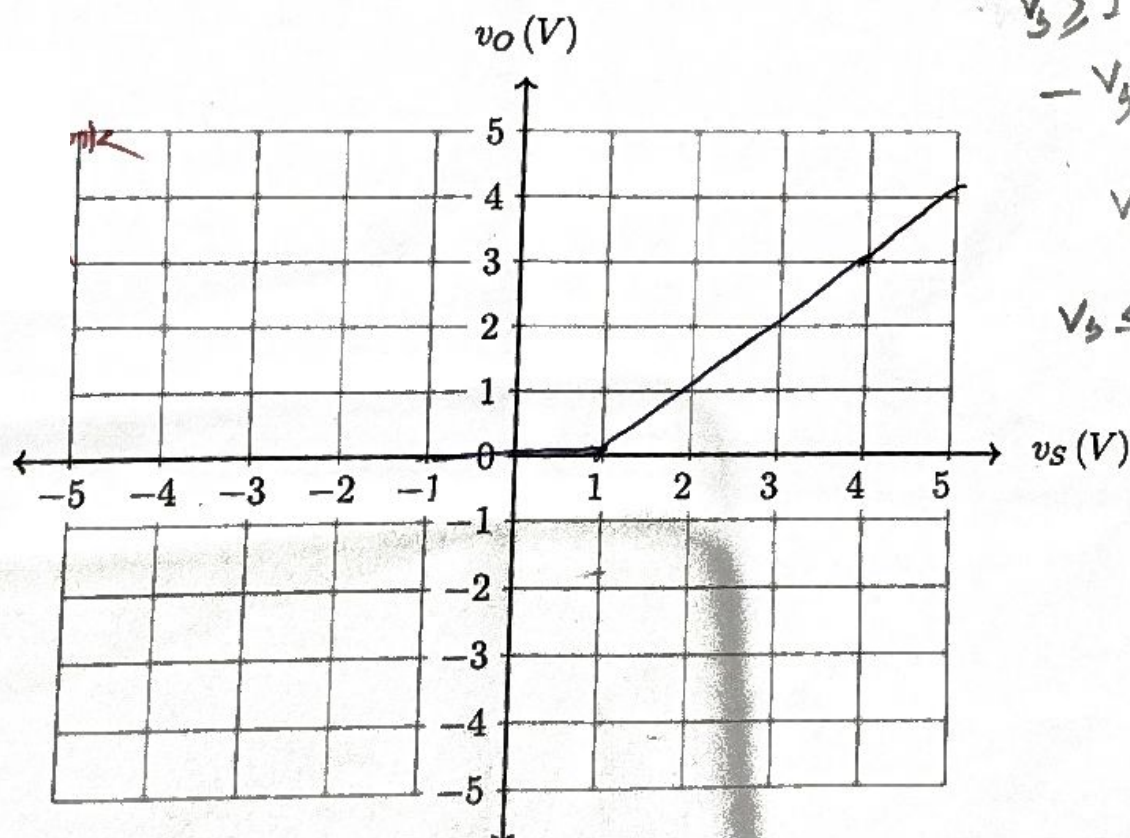
## ■ Question 3 of 5

[CO2] [5 marks]

A rectifier circuit is designed to drive a resistive load  $R_L$  using a diode  $D_1$  with cut-in voltages of 1 V, as shown in the schematic below. The circuit rectifies an AC input voltage  $v_S(t)$ , illustrated in the accompanying plot.



- (a) [2 marks] On the same grid as  $v_S(t)$ , sketch approximately the waveform of the output voltage  $v_O(t)$ .
- (b) [3 marks] Write an equation relating  $v_O(t)$  and  $v_S(t)$ , and draw the voltage transfer characteristic (VTC) graph for the rectifier on the following blank grid.



$$\begin{aligned}
 v_S &\geq 1 \\
 -v_S + v_{D_0} + v_O &= 0 \\
 v_O &= v_S - v_{D_0} \\
 v_S &\leq 1, \quad v_O &= 0
 \end{aligned}$$

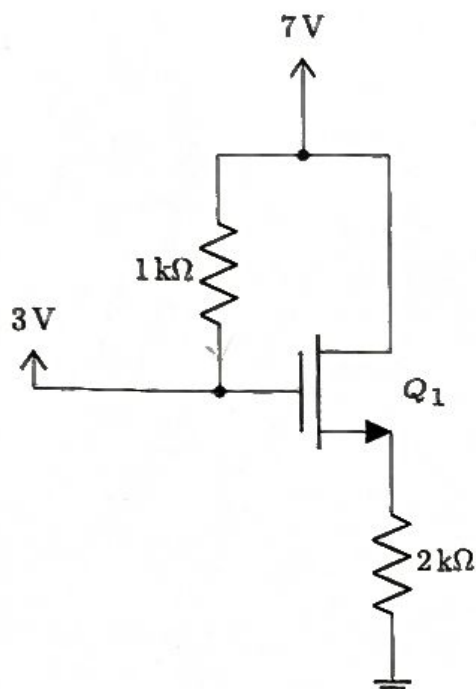


### Question 4 of 5

[CO2] [22 marks]

Analyze the following circuits to answer the questions below:

- (a) [8 marks] The NMOS transistor  $Q_1$  in the following circuit has parameters given in the box. Determine the current supplied by the 7 V source.



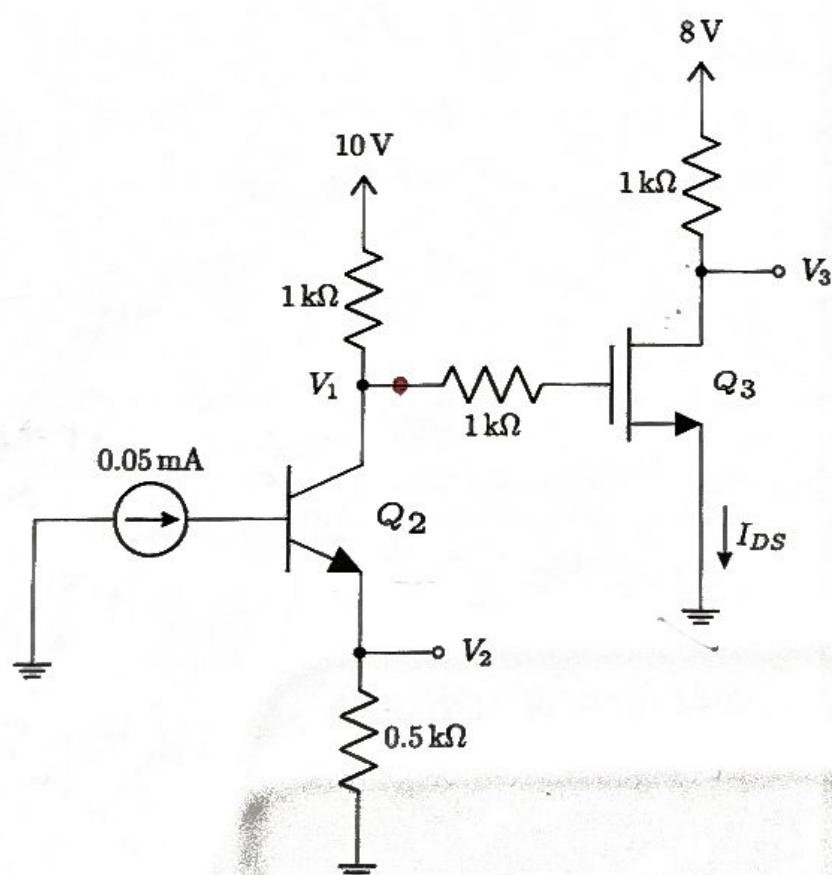
#### N-Channel MOSFET Parameters:

For the transistor  $Q_1$ ,

Threshold voltage:  $V_{T_n} = 2\text{ V}$

Transconductance parameter:  $k_n = 2\text{ mA/V}^2$

- (b) The transistors  $Q_2$  and  $Q_3$  in the following circuit have parameters given in the box.



#### NPN BJT Parameters:

For the transistor  $Q_2$ ,

Common emitter current gain:  $\beta = 100$

Base-emitter voltage in the active region:

$V_{BE(\text{active})} = 0.7\text{ V}$

Base-emitter voltage in the saturation region:

$V_{BE(\text{sat})} = 0.8\text{ V}$

Collector-emitter voltage in the saturation region:

$V_{CE(\text{sat})} = 0.2\text{ V}$

#### N-Channel MOSFET Parameters:

For the transistor  $Q_3$ ,

Threshold voltage:  $V_{T_n} = 1\text{ V}$

Transconductance parameter:  $k_n = 2\text{ mA/V}^2$

- (i) [6 marks] Determine the values of the voltages  $V_1$  and  $V_2$ .

- (ii) [8 marks] Determine the voltage  $V_3$  and the current  $I_{DS}$ .

#### N-Channel MOSFET Current Equations:

In Saturation:  $I_{DS} = \frac{k_n}{2} (V_{GS} - V_{T_n})^2$

In Triode:  $I_{DS} = k_n \left[ (V_{GS} - V_{T_n}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$

where  $I_{DS}$  is the drain-to-source current, and  $V_{GS}$  and  $V_{DS}$  are the gate-to-source and drain-to-source voltages, respectively.

3 In Question

4. (a) Let  $\rightarrow$  saturation mode

$$I_D = \frac{1}{2} k_n (V_{GS} - V_T)^2$$

$$\frac{V_S}{2} = \frac{1}{2} \times 2 (3 - V_S - 2)^2$$

$$\Rightarrow V_S = \textcircled{2}, 0.5$$

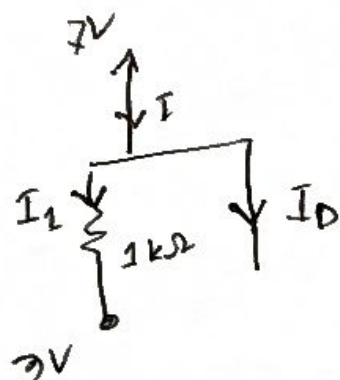
$$V_{GS} - V_S = 3 - 2 = 1V < V_{in}$$

Take  $V_S = 0.5 \Rightarrow$

$$V_{OV} = V_{GS} - V_T = 3 - 0.5 - 2 = 0.5V$$

$$V_{DS} = V_D - V_S = 7 - 3 = 4V$$

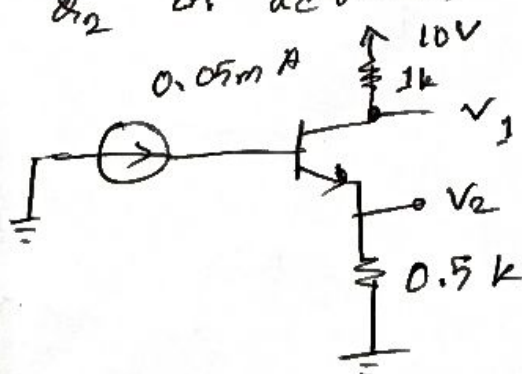
$$V_{DS} > V_{OV} \rightarrow \text{so, mode verified}$$



$$\begin{aligned} I &= I_1 + I_D \\ &= \frac{7-3}{1} + \frac{0.5}{2} \\ &= 4.25 \text{ mA} \end{aligned}$$

$$\text{Power} = 7I = 7 \times 4.25 = 31.5 \text{ mW}$$

(b) No current through  $1k\Omega$  [ $\because I_D = 0$ ]  
let  $Q_2$  in active,



$$I_C = \beta I_B = 5 \text{ mA}$$

$$\frac{10 - V_1}{1} = I_C \Rightarrow V_1 = 5V$$

$$I_E = \frac{V_2}{0.5} \Rightarrow V_2 = 2.525V \quad \left| \begin{aligned} I_E &= I_B + I_C \\ &= 5.05 \end{aligned} \right.$$

Verify?

$$1 \times I_C + V_{CE} + 0.5 \times I_E = 10 - 0$$

$$\Rightarrow V_{CE} = 2.475V > V_{CE, \text{sat}}$$

$\therefore$  verified

Now, for  $Q_3$ ,

$$V_G = V_1 = 5V$$

$$V_S = 0V$$

Assume  $Q_3$  in sat

$$\begin{aligned} I_{D3} &= \frac{1}{2} k_n (V_{GS} - V_T)^2 \\ &= \frac{1}{2} \times 2 (5 - 0 - 1)^2 \\ &= 16 \text{ mA} \end{aligned}$$

$$\frac{8 - V_3}{1} = I_{D3}$$

$$\Rightarrow V_3 = -8 \text{ V}$$

$$\Rightarrow V_D = V_3 = -8 \text{ V}$$

Assumption wrong.

Assume  $Q_3$  in triode

$$I_{D3} = k_n (V_{GS} - V_T - \frac{1}{2} V_{DS}) V_{DS}$$

$$\Rightarrow \frac{8 - V_D}{1} = 2 (5 - 1 - \frac{1}{2} V_D) V_D$$

$$\Rightarrow V_D = 8 \text{ V}, 1 \text{ V}$$

$$V_{DS} = 8 \text{ V}$$

$$\begin{aligned} V_{OV} &= V_{GS} - V_T \\ &= 5 - 0 - 1 \\ &= 4 \text{ V} \end{aligned}$$

$V_{DS} > V_{OV} \rightarrow$  not triode

$$I_{D3} = \frac{8 - 1}{1} = 7 \text{ mA}$$

$$V_{DS} = 1 \text{ V}$$

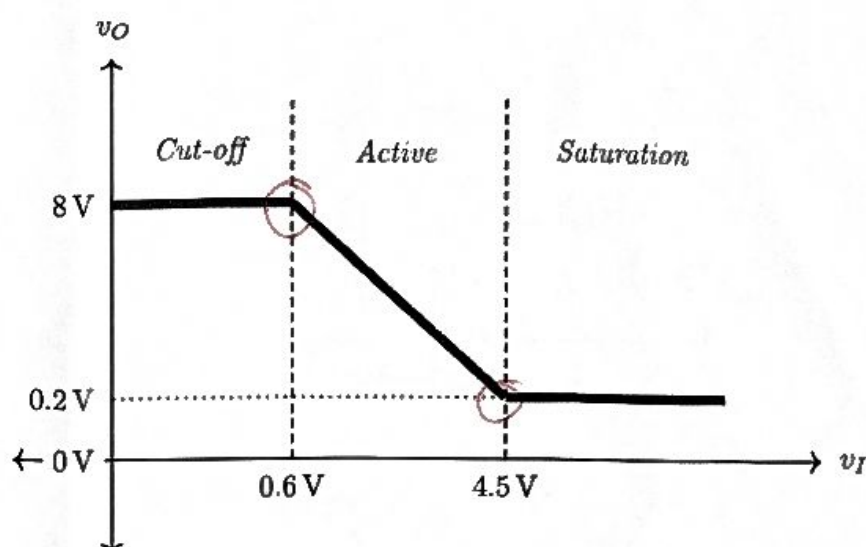
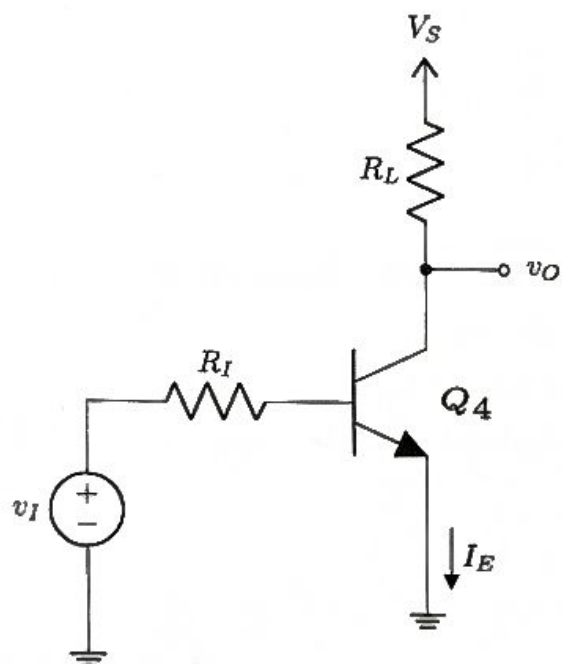
$V_{DS} < V_{OV}$   
Correct!



## ■ Question 5 of 5

[CO2] [10 marks]

The output voltage  $v_O$  for the circuit shown below is plotted as a piecewise function of the input voltage  $v_I$ . The different regions of transistor operation are indicated by vertical dashed lines, and the specific transistor parameters are provided in the accompanying box:

**NPN BJT Parameters:**For the transistor  $Q_4$ ,Common emitter current gain:  $\beta = 100$ Base-emitter voltage:  $V_{BE(sat)} = V_{BE(active)} \approx 0.6\text{ V}$ Collector-emitter voltage:  $V_{CE(sat)} = V_{CE(edge\ of\ sat)} \approx 0.2\text{ V}$ 

Analyze the graph and the circuit to answer the questions below:

- (a) [1 mark] Determine the value of  $V_S$ .
- (b) [6 marks] When the transistor operates at the edge of saturation, the emitter current is  $I_E = 3.94\text{ mA}$ . Determine the values of  $R_L$  and  $R_I$ . [Use  $\beta = 100$ ]
- (c) [3 marks] Using the values found in part (b), what is the value of  $I_E$  when  $v_I = 5\text{ V}$ ?

(a) In cut off mode,  $I_E = I_C = I_B = 0$

$$I_C = \frac{V_S - v_O}{R_L} = 0$$

$$\Rightarrow V_S - v_O = 0 \Rightarrow V_S = v_O$$

$$\Rightarrow V_S = 8\text{ V}$$

$$I_B = \frac{v_I - V_{BE}}{R_I}$$

$$\Rightarrow R_I = \frac{4.5 - 0.6}{0.039} = 100\text{ k}\Omega$$

$$I_B = \frac{I_E}{\beta + 1} = 0.039\text{ mA}$$

$$I_C = I_E - I_B = 3.901\text{ mA}$$

(b) At the edge of saturation,

$$v_I = 4.5\text{ V}$$

$$v_O = v_C = 0.2\text{ V}$$

$$V_{BE} = 0.6$$

$$\Rightarrow v_B - v_E = 0.6$$

$$\Rightarrow v_B = 0.6\text{ V}$$

$$R_L = \frac{V_S - v_O}{I_C} = \frac{8 - 0.2}{3.901} = 2\text{ k}\Omega$$

(c) When,  $v_I = 5\text{ V}$ ,  $v_O = 0.2\text{ V}$

$$I_C = \frac{V_S - v_O}{R_L} = 3.9\text{ mA}$$

$$I_B = \frac{v_I - V_{BE}}{R_I} = \frac{5 - 0.6}{100} = 0.044\text{ mA}$$

$$I_E = I_C + I_B = 3.944\text{ mA}$$

No  
Partial